

MA7444 Excel for Data Science

Audit Trail & Business Report

Coursework 2

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MA7444

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Audit Trail

Overview

The project involves the development of a Microsoft excel-based decision support system to analyse the procurement and spending habits of Birmingham City Council. Using open-source government data, the project aims to transform raw transactional logs into actionable business insights.

Data sources, Checks and Corrections

Data Source:

The analysis is based on public sector transparency data provided by the Birmingham City Council for expenditure over £500.

- Dataset Name: Payments to Suppliers over £500
- Source URL: [Birmingham City Observatory](#)
- Publisher: Birmingham City Council
- Update Frequency: Monthly
- License: Open Government Licence v3.0

Checks and Corrections:

- When importing the data using the Power Query, I made sure to select the filter to remove any rows with an empty value for the Amount variable because it would be irrelevant to include an empty value transaction for this analysis.
- There was an error in the Merchant Category column where one of the merchants namely 'Food & Drink Purchases' had a double spacing in the between the word 'Food' and the character &. To fix this, I used the replace function CTRL + H to replace the double spaces with a single space and this made 88488 corrections. The Beneficiary column also had some of the errors so applied the same fix to all columns in the table.
- Formatted the Amount column to Currency to match the context that the amounts represent money spent.

Processes and Assumptions

- After importing the data, I copied the data and pasted the data in new workbook. I then protected the raw data and continued to edit in the new workbook.
- I intended to categorise the transactions presented in the Amount column into categories, 'Very Low', 'Low', 'Medium', 'High' and 'Very High'. A column was added in the Council_Analysis_Table with the XLOOKUP function using the Spend_Bend_Table in the same sheet applied to all rows.

- Date extraction into separate year and months was done and added as columns at the end of the Council_Analysis_Table using functions such as =YEAR(B2) and Month(B2) to make it easier to process text columns when using pivot tables.
- To calculate the total amount spent in the whole period I used the SUM function on the Amounts column to add all the money spent in all transactions.
- To create a graph that was represented the forecast of the money due to be spent, I used the exponential smoothing algorithm on the historical monthly data to project the future liabilities. This is possible in as a function in excel under the Data tab and selection of Forecast Sheet option on data extracted from a pivot table on a different table.
- To find the total number of invoices processed by all the departments, I found the sum of the number of columns in the whole table containing the database of all transactions in the council using the ROWS function.
- I assumed that there is no maximum in the amount spent on 1 transaction, so I did not check for maximum amounts spent in the amount column. I used the function COUNTIF on the amount column to check if there were any values below.
- To count/ identify the number of Beneficiaries, Directorates, Supplier Numbers, Summaries and Merchant Categories I used the COUNTA and UNIQUE functions.

Worksheets

Cover Sheet: This is the welcome page briefly describing the project and had buttons to other worksheets.

Council_Spend: Raw data imported from the Birmingham City Council website.

Council_Spend_Analysis: Analysed data in a table format with slices to filter the table and a few checks performed in cells next to the table.

Pivot_Analysis: All pivot tables present for analysis.

Spend_Band: The spend band table.

Payment_Forcecast: The table of the data and forecast chart showing projected council expenditure.

Dashboard: An interactive tool used to see the key results from all analysis of the data.

Pivot Tables

Period_Totals: A table showing the sum of money spent per quarter and per year.

Beneficiaries: Top 5 Suppliers each year.

Directorates: Top 5 Directorates each year.

Spend_Band: Sum of money spent within each spending band.

Band_Counts: Number of transactions per spending band per year.

Total_Amounts: Grand Total.

Beneficiary_SupplierNumber: Names of Beneficiaries that represent individual supplier numbers.

Directorate_Beneficiary: Sum and Averages paid by each Directory and the number of Beneficiaries each directory paid.

Directorate__Merchant: Sum of money spent by each directory and top 5 merchant categories they are spending most of their money on.

Directorate_Summary: Sum of money in each directorate and what it is allocated to within the department.

Business Report

Executive Summary

This report presents a data-driven analysis of Birmingham City Council's expenditure between January 1st 2024 to August 29th 2025. The objective is identifying cost patterns, drivers, asset allocation, supplier dependency and all parameters surrounding the use of funds to service the council.

Methodology

The raw expenditure extracted from the source was processed through the Power Query feature in Microsoft Excel. This is where the initial standardisation of the data was done and previewed.

The date column was then split into years and months for easier access to the dates of the transactions when using pivot tables to extract specific data from the database.

A logical approach was also taken to subdivide transactions into specific categories based on their amounts and that gave us an ability to monitor the amounts spent by the council according to specific bands.

A prediction system was also incorporated to assess the future expenditure of the Council for future planning purposes.

Few additional Microsoft Excel sheets were also used to do some computations and finally an interactive dashboard was created to display the key findings of the data.

Results and Observations

Expenditure over time

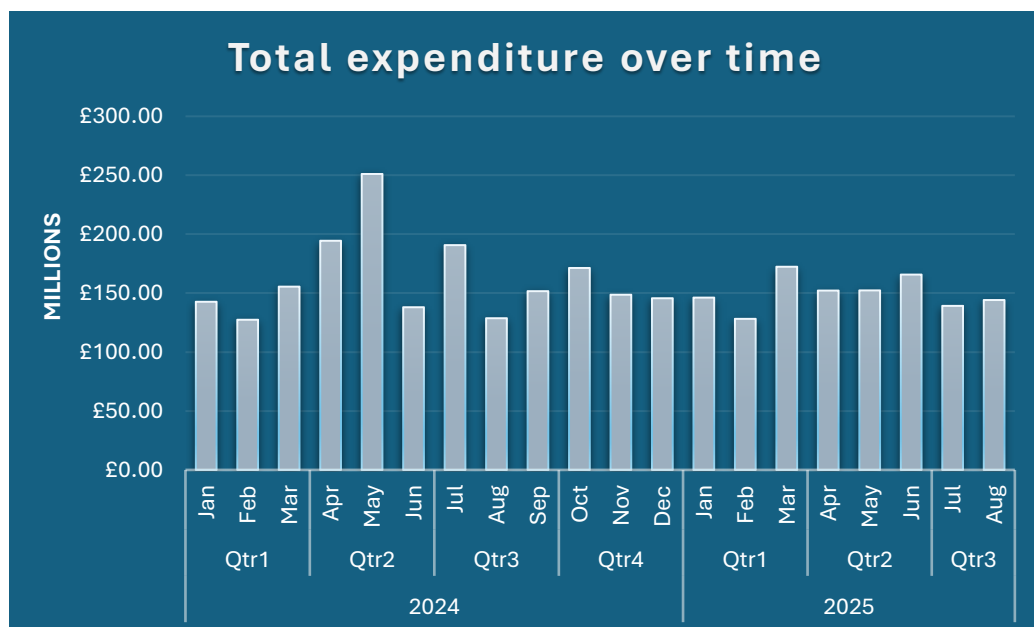


Figure 1

Figure 1 describes the general trend of the city council across different departments for the duration between January 2024 and August 2025. The graph had the amounts of money spent each month and is split in quarters for each year.

As it stands, May 2024 had the highest expenditure at +£250 Million followed by April and July in the same year at around £194 Million. This spike was however not repeated in the following year in 2025 with the highest spending being in March at £174 Million.

The lack of consistency in the data may be due to a difference in the projects that are being undertaken at different times of the year, but they give a good ability to understand that different financial periods may require different budgets depending on the priority of their projects.

Row Labels	Sum of Amount
2024	
Qtr1	
Jan	£142,681,293.02
Feb	£127,429,274.13
Mar	£155,514,913.55
Qtr2	
Apr	£194,453,557.30
May	£251,105,700.89
Jun	£138,039,705.07
Qtr3	
Jul	£190,731,561.85
Aug	£128,703,115.20
Sep	£151,656,685.10
Qtr4	
Oct	£171,351,301.98
Nov	£148,645,649.57
Dec	£145,661,056.36
2025	
Qtr1	
Jan	£146,201,587.68
Feb	£128,194,239.09
Mar	£172,395,029.84
Qtr2	
Apr	£152,178,806.11
May	£152,319,964.07
Jun	£165,739,472.96
Qtr3	
Jul	£139,182,576.67
Aug	£144,147,915.42
Grand Total	£3,146,333,405.86

Figure 2

Figure 2 Displays the total spendings of the Birmingham City council per month, quarter and year with a grand total of about £3.15 Billion. 2024 naturally contributes the highest amount of money to the grand total because of its complete data for all the months. A comparison could

be made by months and quarters to analyse further which year had the most and least expenditure and by the looks of the figures, 2025 had less spendings.

Expenditure by Directorate

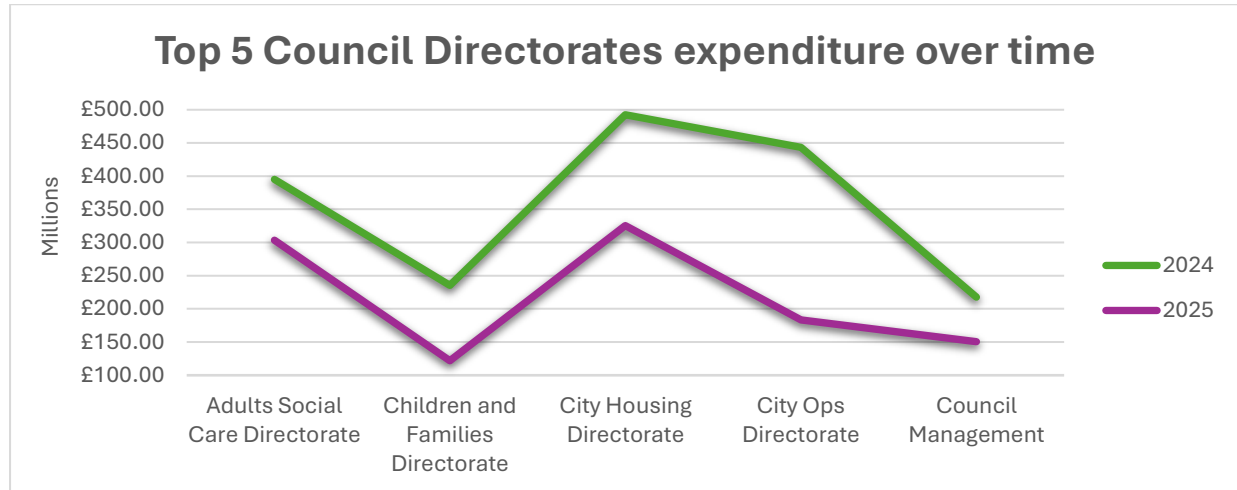


Figure 3

Figure 3 showcases the top 5 council directorates based on their expenditure over the two years. They all rank highly as the top five within the council for both years in a similar pattern. Based on the graph, City Housing Directorate has the highest expenditure in both years and totals just under £818 Million with Adults Social Care Directorate following suite in the totals at just under £700 Million.

Directorates	Sum of Amount	Average of Amount	Count of Beneficiary
City Housing Directorate	£817,871,251.17	£57,861.43	14135
Adults Social Care Directorate	£698,436,442.22	£7,837.56	89114
City Ops Directorate	£626,613,461.43	£17,434.99	35940
Council Management	£368,132,424.69	£19,497.51	18881
Children and Families Directorate	£357,264,524.41	£7,747.92	46111
Places, Prosperity & Sustainability Directorate	£131,409,981.08	£17,920.36	7333
Strategy, Equalities & Partnerships	£126,908,540.14	£146,715.08	865
HR, OD and Transformation	£12,121,557.79	£202,025.96	60
Finance	£5,997,832.60	£11,556.52	519
Strategy, Insight and Governance Deputy CE	£1,007,724.04	£5,167.82	195
Finance Control- Monitoring	£391,300.60	£48,912.58	8
Finance Control- Other	£122,059.61	£40,686.54	3
Birmingham City Council Directorate Dir	£48,540.08	£9,708.02	5
Non-BCC N2 - Finance Control- Other	£7,264.00	£3,632.00	2
Non-BCC N2 - Adults Social Care Dir	£502.00	£502.00	1
Grand Total	£3,146,333,405.86	£14,759.60	213172

Figure 4

Figure 4 is a table showing the total amounts spent by all directories, the average amounts spent by the directories and the number of transactions or number of beneficiaries they had to pay in the timeline. This data allows us to investigate the question, Which departments are creating the most administrative paperwork? Processing an invoice costs money which

translates to admin time and a department processing thousands of tiny invoices is inefficient compared to one processing a few large ones if the monetary payment is a low sum. A high administrative burden may cause suggestions such as a directorate using Purchase Cards instead of formal invoices to process transactions with the suppliers.

Supplier Dependency and Top Beneficiaries

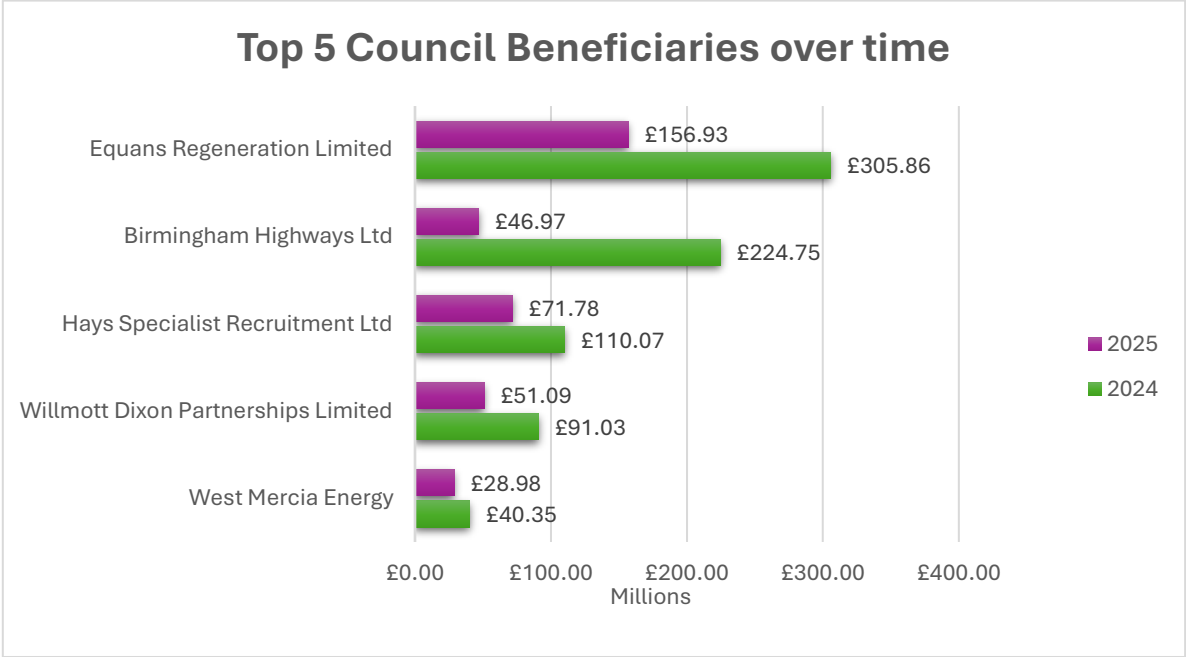


Figure 5

Figure 5 highlights the top 5 Beneficiaries of the council in both years. Beneficiaries who have their 2024 and 2025 figures close in amounts could help indicate at what time of the year the services are required by the council.

There is an average number of 6000 suppliers counted and Equans Regeneration Limited comes out on top in terms of money received for the products or services they provide for the council in both years. This highlights a strong relationship between the council and the company and naturally would create a risk that if there was any fault in the project, it would negatively impact the council at a major scale. More care should be taken to monitor the projects because any slight delays may reflect in a monetary value for the council.

Transaction size and Audit Priority

Spend_Band_Table

Spend_Thereshold	Spend_Band
0	Very Low
1000	Low
5000	Medium
15000	High
50000	Very High

Spend band table used to create thresholds for money spent on transactions and a colimn in the main analysis table

Figure 6

Figure 6 describes the thresholds that were used to create spend bands for the council. These will be used to analyse totals for each band and the frequency of the transactions.

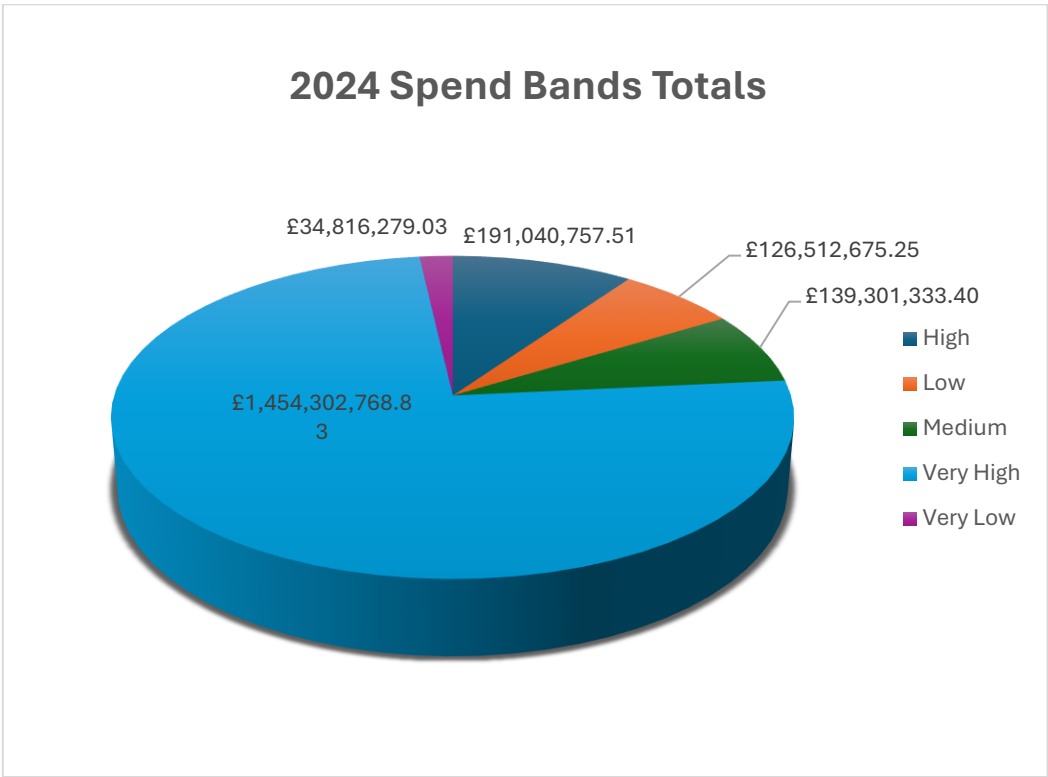


Figure 7

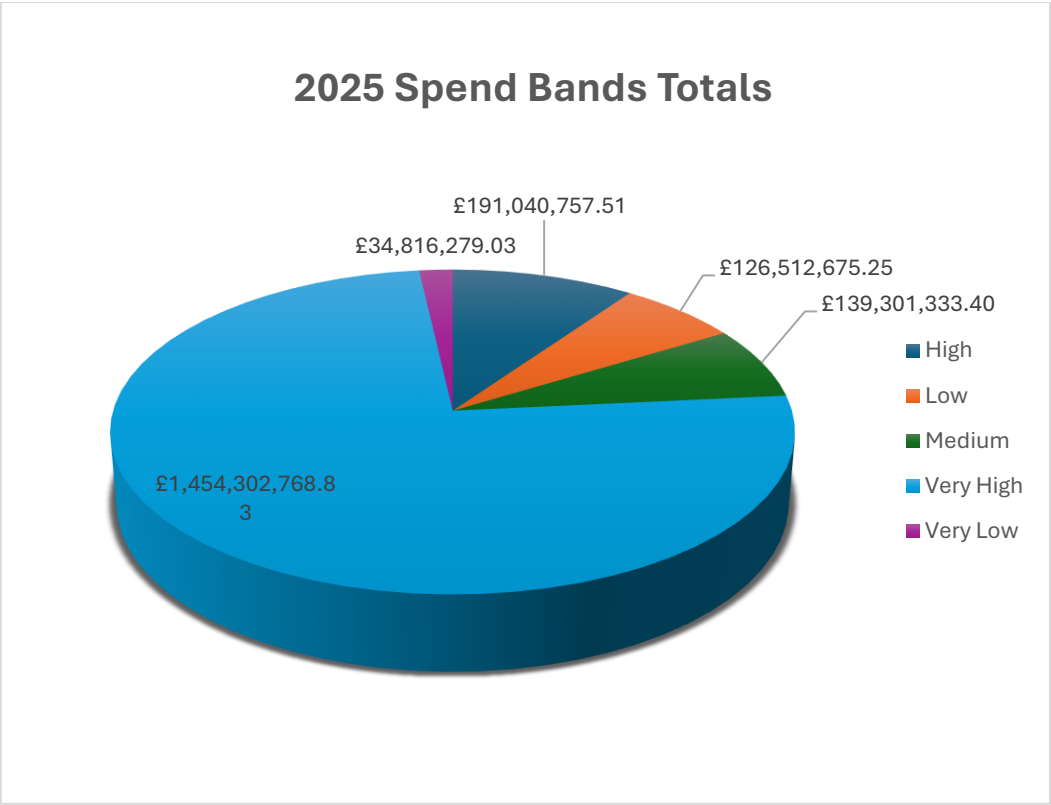


Figure 8

Figure 7 and Figure 8 show the amount of money spent each year in the specific spending bands.

Spend Band			
	2024	2025	
High	£191,040,757.51	£126,304,015.29	
Low	£126,512,675.25	£76,625,162.26	
Medium	£139,301,333.40	£85,996,449.69	
Very High	£1,454,302,768.83	£892,385,179.63	
Very Low	£34,816,279.03	£19,048,784.97	
Grand Total	£1,945,973,814.02	£1,200,359,591.84	
Count of Spend_Band Column Labels			
Row Labels	2024	2025	Grand Total
High	7038	4689	11727
Low	57823	36314	94137
Medium	16419	9990	26409
Very High	3437	2555	5992
Very Low	48580	26327	74907
Grand Total	133297	79875	213172

Figure 9

Figure 9 displays two tables, one with the total spent in each spend band and each year while the other table shows the number of transactions each spend band had within each year. As the spending band increases, there was an increase in the money spent in every year. That means that low value transactions added up to give low totals and high value transactions added up to give high totals. The table below gives an indication of the frequency of the transactions and shows that Low transactions accounted for most of the transactions over all categories while the Very High transactions were the least. This gives an insight on the type of transactions that the council processes the most and the least giving the management a direction to investigate categorically based on their priority in cutting costs.

What is being bought?

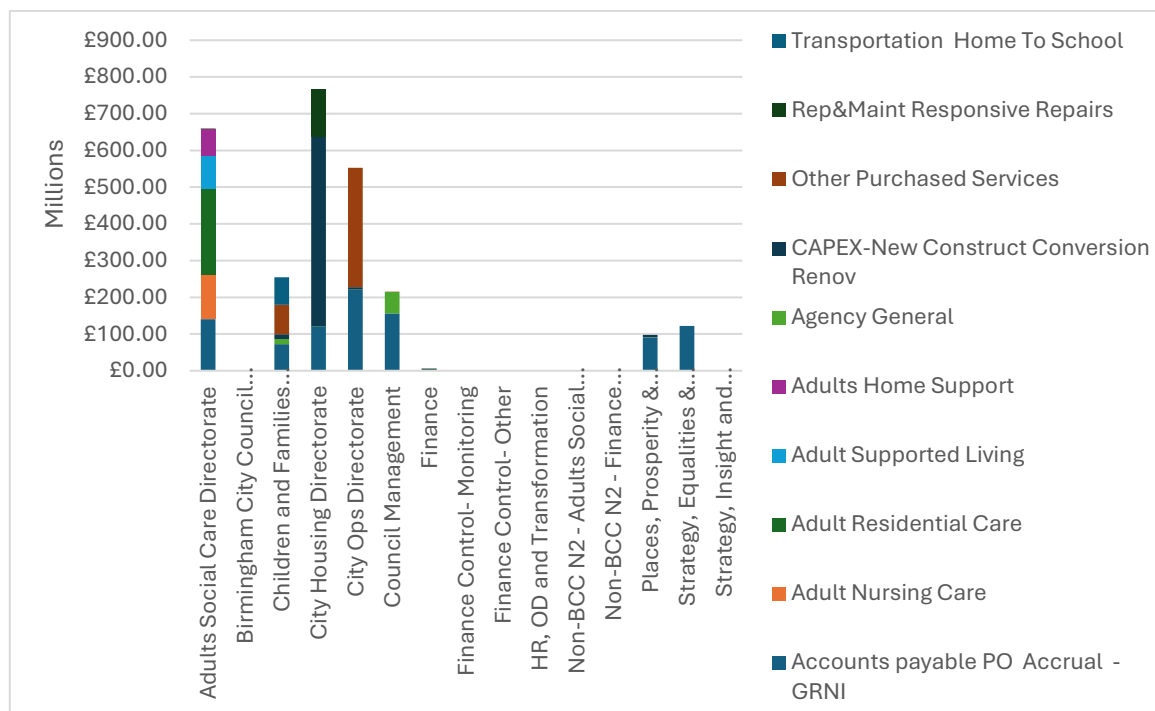


Figure 10

To answer the question of what is actually being bought, Figure 10 gives insight into the Directorate_Merchant pivot table. It shows what directorates allocate their budgets to buying. Each merchant category briefly gives an idea to the priorities of the directorate, and how much money is being spent in comparison to other categories. From the graph we can see that the highest priority of the City Ops Directorate is the CAPEX-New Construct Conversion Renovation and it amounts to about £600 Million compared to the Rep&Maint Responsive Repairs which amount to just over £100 Million.

The expenditure of the Adults Social Care Directorate is spread out more equally between all the merchant categories. The Directorate_Summary pivot table also allows us to identify even within the directorate what specific type of transaction would be consuming the budget on a larger scale compared to other transactions and allows analysis on different procurement risks.

Financial forecast

The model in Figure 11 forecasts an average of £144 Million for the months of September, October, November, December 2025 and January 2026. This prediction model used the

previous year historical data to give the estimates for each month at a 95% confidence interval and has the error bars representing the possibility of any fluctuations that may occur. In this case we can state that majority of the transactions should fall between £128 Million and £214 Million.

This information allows the Council Board to assess the budget they had allocated and possibly improve liquidity planning as the months go by. Any money saved from this time could be reserved to be allocated to other months such as May that have seasonal spikes.

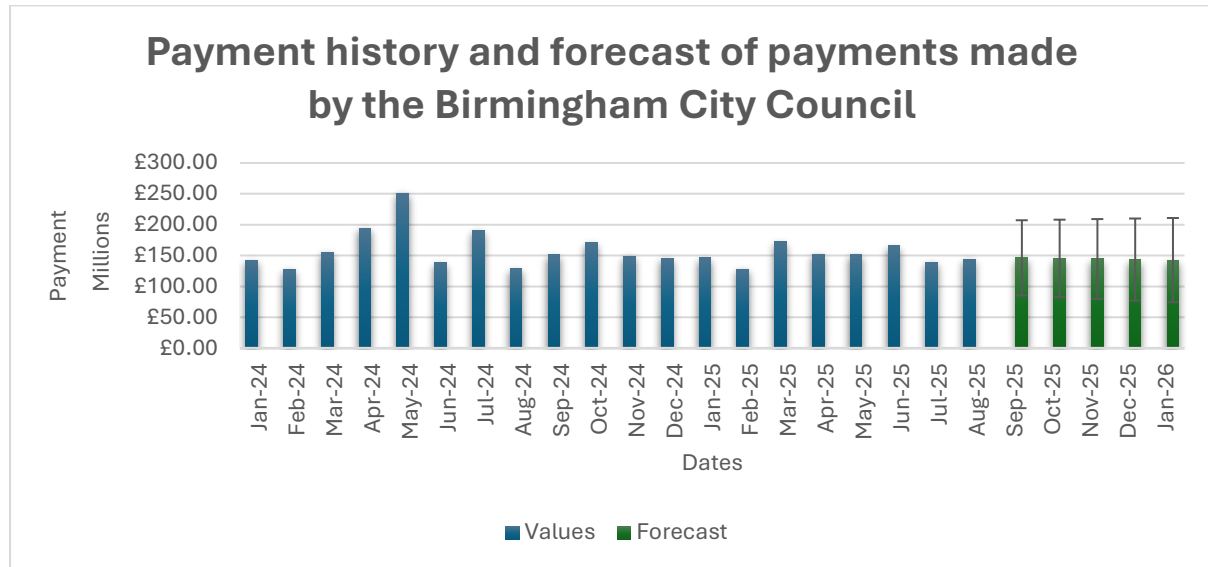


Figure 11

Data mismatch

The only occurrence of missing data is represented in the mismatch between the number of Beneficiaries and the Supplier Number. It is mentioned however in the data source that some information falls under the commercially sensitive title such as payments relating to vulnerable young people or adults. The supplier numbers of such transactions were represented under the Beneficiary names Redacted Personal Data, Redacted - Personal Data and Redacted-Personal Data as shown in the Figure 12 below.

REDACTED - PERSONAL DATA
B10068
B101249
B10242
B103163
B103211
B103495
B105171
B105562
B105822
B105908
B106414
B10865
B110179
B113886
B115812
B116181
B116270
B11891

Figure 12

Conclusions

The analysis confirms that while the Council is spending a large amount of money, it seems to be concentrated in certain areas. There is a strong connection to the performance of the City Housing Directorate and the stability of Equans Regeneration Limited. In terms of operational strain, the volume of low value transactions in the Adults Social Care directorate needs to be revised.

Key Findings:

Total expenditure: amount spent by Birmingham City Council is £3,146,333,405.86

Highest spending directorate is the City Housing Directorate at £817,871,251.17

Largest beneficiary is Equans Regeneration Limited at £462,795,612.98

Future Steps and Recommendations

Audit Focus: The internal audit should firstly focus extensively on transactions and directorates that have majority of their payments in the very high and high spending band. After all transactions are cleared then the next focus could be into high administration directorates which are represented by ones that have the highest number of transactions as this is where the greatest number of fraudulent activities may occur.

Risk Mitigation: Develop contingency plans for the top 5 beneficiaries due to the high level of financial exposure to the suppliers. It may be worthwhile to come up with ways to diversify and decentralise the reliance on them.

References

Birmingham City Observatory (2025). *Payments to Suppliers over £500*, [Birmingham City Observatory](#)